*log(x)* -- returns the natural logarithm of x
*sin(x)* -- returns the sine of x
*sqrt(x)* -- returns the square root of x
Here is an example:

```
{
  x = 5.3241
  y = int(x)
  printf "truncated value is ",  y
}
Output: truncated value is 5
```

## String functions in AWK

1. *length(string):* Calculates the length of a string.
2. *index(string, substring):* Returns the first position of the substring within a string. For example, in *x= index("programming", "gra")*, x returns the value 4.
3. *substr():* Extracts the substring from a string. The two different ways to use it are: *substr(string, position)* and *substr(string, position, length).*
   Here is an example:

```
{
  x = substr("methodology",3)
  y = substr("methodology",5,4)
  print "substring starts at " x
  print " substring of length " y
}
```

The output of the above code is:

```
Substring starts at hodology
Substring of length dolo
```

4. *sub(regex, replacement string, input string)* or *gsub(regex, replacement string, input string)*
   *sub(/Ben/," Ann ", $0):* Replaces Ben with Ann (first occurrence only).
   *gsub(/is/," was ", $0):* Replaces all occurrences of 'is' with 'was'.
5. *match(string, regex)*

```
{
    x=match($0,/^[0-9]/)              #find all lines that
start with digit
    if(x>0)              #x returns a value > 0 if there's a
match
      print $0
    }
```

6. *toupper()* and *tolower():* This is used for convenient conversions of case, as follows:

```
{
print toupper($0)      #converts entire file to uppercase
}
```

## User defined functions in AWK

AWK allows us to define our own functions, which enables reusability of the code. A large program can be divided into functions, and each one can be written/tested independently.

```
The syntax is:
    Function Function_name (parameter list)
    {
        Function body
    }
```

The following example finds the largest of two numbers. The program's name is *awfunct.*

```
{
print large($1,$2)
}
function large(m,n)
{
    return m>n ? m : n
}

Input file is: doc
          100 400
Running the script: awk -f awfunct doc
          400
```

## Associative arrays

AWK allows one-dimensional arrays, and their size and elements need not be declared. An array index can be a number or a string.

The syntax is:

```
arrayName[index] = value
```

Index entry is associated with an array element, so AWK arrays are known as associative arrays.

For example, *dept[$2]* indicates an element in the second column of the file and is stored in the array, *dept*.

```
    Program name : awkarray
BEGIN{print "Eg of arrays in awk"}
{
    dept[$2]
    for (x in dept)
    {
      print a[x]
```